Adaptive Reading Budgets for Joint Forecast Audits with Cumulative Data

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Abstract

Background. Model audits share verification readings across several predictors. We ask how many readings suffice for precise comparisons and when choosing later readings after earlier answers reduces this requirement.

Methods. We study sharp intervals for affine model-loss contrasts on compact convex sets using a finite query dictionary. We prove fixed-budget bounds, compute minimum fixed reading budgets, and replay certified policies on cumulative-meter data.

Results. For one comparison, if a deterministic policy guarantees precision with two exact scalar readings, some randomized fixed batch succeeds with worst-case probability at least $1/2$. This lower bound disappears with two comparisons. For every $m \geq 3$, we construct instances where adaptation always succeeds but the best randomized batch succeeds with worst-case probability exactly $1/m$. All coordinates are queryable, the budget remains two, and each comparison is separately certifiable by a fixed batch. The separation also holds for all pairwise comparisons among three models. On 225 of 1,098 meter blocks across six systems, two adaptive readings guarantee halving every comparison's width, while the minimum fixed budget is three. Of these, 222 have optimal randomized two-reading batch success exactly $1/2$. The one-third reading reduction on qualifying blocks becomes 6.40% and 7.67% across the complete original and external panels. Capacity information reduces the advantage, and measured public-API retrieval favors bulk.

Conclusions. Sharing a reading budget can make feedback necessary for assurance even when separate comparisons need none. The construction proves a fixed-budget boundary; the retrospective meter study quantifies its narrower benefit in required readings, separately from retrieval cost.

Keywords: artificial intelligence; machine learning; adaptive information; minimax risk; optimal recovery; robust optimization.

1 Introduction

A model audit may need more verification data before it can report a precise comparison. The auditor can retrieve a complete record, choose a batch of readings, or choose later readings after

seeing earlier answers. We study this last option, adaptive querying. Our output is a sharp interval: the smallest interval containing the loss difference for every record compatible with the observations. Its width measures how much the reported comparison can still change. We ask when feedback is needed to meet a stated width requirement with the same reading budget. This is an acquisition problem for fixed predictors. The auditor specifies which loss contrasts must be precise, what information is already available, and which measurements may be requested. We first compute the smallest sufficient fixed reading budget, then test whether two adaptive readings can reduce it. The algorithm returns a reading plan and checks its intervals. Identifying a better model is a separate condition: a comparison interval must also exclude zero.

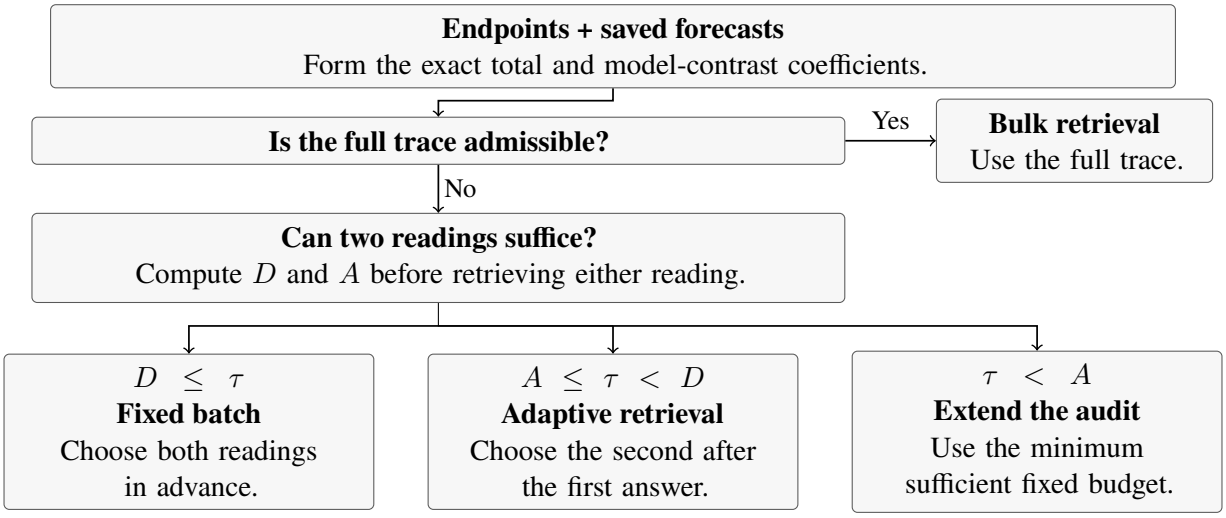


Figure 1: Choosing an audit protocol. For the exact-total, noiseless prefix model, let $J_Q = \max_r W_{r,Q}/W_{r,0}$ over contrasts with positive initial width. Let $D = \min_Q \sup_y J_Q(y)$ and $A = \min_{P \text{ det. adaptive}} \sup_y J_{Q_P(y)}(y)$ under the same two-reading budget. We fix the target τ before retrieval; the main application uses $\tau = 1/2$. Since $A \leq D$, the three restricted-access branches exhaust the possibilities for two-reading certainty, including randomized policies (Proposition 2). Their optimal success probabilities below one are assessed separately. The fixed-budget calculation in Section 2.5.2 supplies an explicit fallback when two readings cannot guarantee success. Bulk is preferred on the measured public API when its full response is admissible (Section 3.2).

The central result is a boundary between one comparison and a shared collection of comparisons. For one affine contrast, a deterministic policy that guarantees precision with two exact scalar readings implies a randomized fixed batch with worst-case success at least one half (Theorem 1). With two contrasts, that floor disappears. For every $m \geq 3$, Theorem 2 constructs instances with optimal adaptive success 1 and optimal batch success $1/m$, still using two readings. Each contrast separately has a successful fixed batch; all coordinates are queryable at equal cost. The same separation holds for all pairs among three nonnegative forecast vectors (Corollary 1). The dictionary grows with m, and three fixed queries suffice. Moreover, the optimal batch mixture is unique: it assigns mass $1/m$ to each of m necessary batches. Even positive worst-record success requires all m batches. This exact support requirement isolates what fails when separately sufficient measurements share a budget.

We make that distinction computable for cumulative-meter audits. An exact two-query planner and matching certificates identify 165 of 819 original blocks and 57 of 279 external blocks where feedback guarantees precision for all six model pairs, while the best randomized batch succeeds with worst-case probability exactly one half. Complete-panel replay then measures what the policies achieve on the stored answers. We also determine the minimum fixed reading budget: 168 original and 57 external blocks require three fixed readings but only two adaptive readings. Thus the application measures the readings needed for assurance. We compare this requirement with the retrieval interface in a separate public-API control.

The same calculation can rule out an insufficient budget. Explicit adverse records show that randomizing an adaptive policy cannot restore certainty on any of the 583 blocks without a deterministic guarantee (Proposition 2). Thus the method supports a three-way decision: choose a fixed batch, use feedback, or obtain more information.

The paper develops the scalar bound, the shared-budget construction and the meter planner, then reports its certificates, recorded outcomes and controls. The retrospective study reuses frozen source panels. Online Resource 1 retains the supporting scalar penalty classification (S11), a separate mean-joint-width construction (S14), historical controls and executable source-to-table verification.

1.1 Related work and the contribution boundary

Partial identification studies conclusions that survive incomplete observations (Manski, 2005; Maia Polo et al., 2024). Cancellation in paired squared losses already yields an affine function of the missing truth (Patton, 2011). Active evaluation and model selection acquire labels to estimate performance or choose predictors (Sawade et al., 2012; Hara et al., 2024; Kossen et al., 2021; Okanovic et al., 2025). We instead require the entire compatible interval of each fixed-record contrast to meet a declared precision target. Kuboyama (2026, Theorem 1 and Proposition 1) studies completion-based model-choice certificates with adaptive binary labels and shared paired bounds. Its distinction between hindsight certificate size and acquisition requirements (Theorem 2 and Proposition 3) is also relevant here; our guarantees concern joint widths for continuous targets.

The width of an information fiber is the established local diameter of information (Novak and Woźniakowski, 2008, Section 4.1). Our shared objective is its weighted ℓ_∞ version for a vector target. Classical deterministic adaptivity bounds cover linear information on convex symmetric classes (Novak and Woźniakowski, 2008, Theorems 4.4–4.5); nonsymmetric recovery is studied by Novak (1995), and noisy scalar criteria by Osipenko (2019, Theorems 2–3 and 5). Modern randomized recovery minimizes reconstruction error (Krieg et al., 2025, Section 2 and Proposition 4.1) (Krieg and Ullrich, 2026, Section 9.1, Eq. (9.15), author manuscript v2). Our precision objective instead asks whether the full terminal fiber is sufficiently narrow. For supporting mean-width results, the order of the worst-record and randomization operations also differs from expected global radius (Krieg and Ullrich, 2026, Eq. (2.4), Remark 6.1, author manuscript v2). Online Resource 1, S11 gives that comparison and the scalar penalty classification, whose sufficient direction applies Sion's minimax theorem (Sion, 1958, Theorem 3.4) and Jensen's inequality.

Information-dependent decisions are established in adjustable robust optimization (Wei and Zhang, 2024; Vayanos et al., 2026). Our scalar precision bound adapts the sufficient-policy result of Kurtz (2026, Theorem 2 and Example 5). Kurtz's Theorem 11 also permits policy counts to

grow with recourse-region complexity. In our construction, each scalar conditional width is affine, but their joint maximum loses the concavity used by the scalar bound. We realize that obstruction with actual coordinate-query fibers: two affine targets and two unit-cost readings have exact success values 1 and $1/m$, although each target alone has a sufficient fixed batch. The adverse records further force support m for any positive batch success and uniquely determine the optimal mixture. The output dimension and budget stay fixed while the measurement dictionary grows. This is the specific contribution beyond general policy-count bounds and the established benefits of adaptive vector recovery.

Adaptive cumulative sampling has close precedents. Gaillard et al. (2023, Section 4.1.1, Corollary 1 and Theorem 3) studies certified scalar integral estimation from a monotone function. For monotone contrast coefficients, summation by parts relates our meter problem to weighted monotone integration. We pair an exact shared-query planner with continuum covering and adverse-record certificates, so each reported precision separation has matching bounds under one measurement interface. The separate aggregate solver in Online Resource 1, S5/S7 builds on weighted-oracle randomization (Kawase and Sumita, 2024), dynamic-program LP representations (Martin et al., 1990), and refinement of tied robust solutions (Iancu and Trichakis, 2014); it is supporting computation rather than the central shared-query contribution.

2 Methods

2.1 Model Comparisons and Sharp Information

Let $a, b \in \mathbb{R}^N$ be two fixed prediction vectors for the same scheduled targets. Their paired mean-squared-error difference is

$$D(y) = \frac{1}{N}\sum_{i=1}^{N}\{(a_i - y_i)^2 - (b_i - y_i)^2\} = d + c^\top y, \qquad c_i = \frac{2(b_i - a_i)}{N}. \tag{1}$$

All scheduled targets remain in the denominator. Initially observed values, documented aggregate measurements and other defensible constraints define the feasible set F. Any permitted archive record that directly reconstructs a target is incorporated before this set is declared. Sharpness is always relative to that declared information set.

The attainable interval of D on F is an identification interval for this record, not a sampling confidence interval. A negative upper endpoint certifies that a has smaller squared loss for every completion; a positive lower endpoint certifies b. An interval containing zero cannot certify a strict winner. Subsequent exact measurements restrict F to a fiber and therefore cannot widen its sharp interval. A prescribed width threshold instead asks for a precision guarantee, regardless of the interval's location.

The mathematical examples below also describe paired squared-loss widths. Given any contrast coefficient c, nonnegative predictions $a_i = N\max(-c_i, 0)/2$ and $b_i = N\max(c_i, 0)/2$ realize that coefficient in (1). The induced constant changes the interval's position while preserving its width. This gives the constructions a paired model-comparison interpretation.

2.2 Information and precision

Let $F \subset \mathbb{R}^n$ be nonempty, compact and convex, and let $T(y) = c^\top y + d$ be the scalar contrast. We select exact linear measurements from a fixed finite dictionary. The nonempty family $\mathcal{Q}$ of legal final query sets is fixed before observing answers and is the same for batch and adaptive policies. For example, fixed costs $k_j \geq 0$ and a hard budget B require $\sum_{j \in Q} k_j \leq B$ on every execution. Write L_Q for the matrix of measurements in Q.

For a feasible record y, the compatible records and sharp width are

$$F_Q(y) = \{z \in F : L_Q z = L_Q y\}, \qquad W_Q(y) = \max_{z \in F_Q(y)} T(z) - \min_{z \in F_Q(y)} T(z). \tag{2}$$

Compactness guarantees both endpoints. This width is the scalar local diameter of information; additive constants in T do not affect it. A randomized batch chooses Q before seeing answers. An adaptive policy uses previous answers and an independent random seed to choose later measurements and when to stop. Query and stopping rules are jointly measurable, and every execution ends after finitely many queries in $\mathcal{Q}$. The hidden record is fixed before the seed is drawn. For bounded-error models, one fixed error vector is part of that record throughout the execution; the resulting feasible set must remain convex.

For either randomized policy class $\mathcal{C}$, define precision success by

$$S_\tau^{\mathcal{C}} = \sup_{\mathcal{P} \in \mathcal{C}} \inf_{y \in F} \Pr_{\omega}\{W_{Q_{\mathcal{P}}(y,\omega)}(y) \leq \tau\}, \qquad \tau \geq 0. \tag{3}$$

Probability concerns the policy's seed at a fixed record; equality at the threshold counts as success. A narrow interval need not exclude zero and therefore need not identify a winning model.

Proposition 1 (Width and the transcript fiber)**.** *For fixed Q, W_Q is finite, nonnegative, concave and upper semicontinuous on F. For a fixed policy seed, the records consistent with an adaptive transcript are exactly $F_Q(y)$ for its terminal set Q.*

Proof. Convex combinations of maximizing/minimizing endpoint pairs are feasible at the corresponding combined record, proving concavity. Along a convergent sequence of records, compactness supplies convergent endpoint subsequences; their limits are feasible in the limiting fiber, proving upper semicontinuity. Thus the widths are Borel measurable. For a fixed seed, records agreeing on queried answers produce the same next query and stopping decision, by induction. Conversely, a consistent transcript requires those answers. The argument includes a transcript with no queries. □

2.3 The scalar two-query precision floor

The following scalar specialization adapts sparse-policy reasoning (Kurtz, 2026, Theorem 2 and Example 5). Online Resource 1, S12 gives the general first-answer-dimension bound and its proof.

Theorem 1 (A half-success guarantee for scalar precision)**.** *Under Section 2.2, suppose a deterministic policy using at most two exact scalar measurements guarantees $W \leq \tau$ for every record. The eligible second measurements are fixed before the first answer, and every resulting final set obeys the common hard budget. Then a randomized nonadaptive policy supported on at most two legal batches has worst-case precision success at least $1/2$.*

Proof. If the policy makes no query, the empty batch succeeds. Otherwise its first measurement $L_0 y$ is fixed. Its feasible answers form a compact interval $I = L_0 F$, possibly a singleton. Index the finitely many legal second actions by j, including an empty action when stopping after the first query is legal. Let

$$g_j(t) = \max\{c^\top (u - v) : u, v \in F,\ L_0 u = L_0 v = t,\ L_j u = L_j v\}.$$

This is the largest terminal width after choosing action j at first answer t: each feasible endpoint pair belongs to a common terminal fiber, and every such fiber supplies feasible endpoints. Convex combinations of endpoint pairs show that g_j is concave. Hence its strict failure set $H_j = \{t \in I : g_j(t) > \tau\}$ is convex. The policy's guarantee implies $\bigcap_j H_j = \varnothing$. Finite Helly's theorem in dimension at most one supplies at most two failure sets with empty intersection (Amenta et al., 2016, Theorem 1.1). This form does not require closed sets, so equality at the precision threshold is preserved. Choose uniformly among the corresponding fixed batches (L_0, L_j). At every record, at least one selected batch has width at most $g_j(L_0 y) \le \tau$. Its success probability is therefore at least $1/2$ (or one if a single batch suffices). □

2.4 A fixed-budget boundary between scalar and joint precision

The scalar half-success floor need not survive adding a second target. The following construction keeps the query budget at two and makes every coordinate available at unit cost.

Theorem 2 (Two-target precision with a fixed two-query budget)**.** *For every integer $m \ge 3$, let*

$$F_m = \{(t, u, v, z_1, \dots, z_m) : 0 \le u \le t \le 1,\ 0 \le v \le 1 - t,\ 0 \le z_j \le 1\}.$$

All $m + 3$ coordinates are exact unit-cost queries. Define

$$a_j = j + \tfrac{1}{4}, \quad b_j = m + 1 - j + \tfrac{1}{4}, \quad H = \frac{m(2m+3)}{4}, \quad M = H - m,$$

and the affine targets

$$T_1 = Mt + mu + \sum_j a_j z_j, \qquad T_2 = M(1 - t) + mv + \sum_j b_j z_j.$$

Both initial target intervals are $[0, 2H]$. With at most two queries on every execution, let success mean that both terminal sharp widths are at most H. Then the optimal worst-record precision-success probabilities are

$$S^\star_{\text{adaptive}} = 1, \qquad S^\star_{\text{randomized batch}} = \frac{1}{m}.$$

Each scalar target separately admits a fixed two-query policy with sure success. The adversarial record is fixed before any policy randomization.

Proof. The polytope F_m is nonempty, compact and convex. Since $\sum_j a_j = \sum_j b_j = H$ and $M + m = H$, the extrema of T_1 are 0 and $2H$, attained at $(t, u, z) = (0, 0, 0)$ and $(1, 1, \mathbf{1})$. The corresponding choices $(t, v, z) = (1, 0, 0)$ and $(0, 1, \mathbf{1})$ give the same extrema for T_2.

Query t, choose the smallest index j satisfying $(j-1)/m \le t \le j/m$, and query z_j. The exact terminal widths are

$$W_1 = H + mt - a_j, \qquad W_2 = H + m(1-t) - b_j. \tag{4}$$

Indeed, conditioning on t leaves $u \in [0,t]$ and $v \in [0,1-t]$; the unqueried z coordinates remain an independent box. The observed value of z_j contributes only a constant. The chosen branch gives $\max(W_1, W_2) \le H - 1/4$, proving adaptive success one.

For the randomized-batch lower bound, choose j uniformly before any observations and query $\{t, z_j\}$. Every feasible t belongs to at least one branch interval, so success probability is at least $1/m$ for every record. This lower bound holds over the full continuum of F_m.

For the upper bound, give equal adverse mass to the m feasible records

$$x_i = \big((i - \tfrac{1}{2})/m, 0, 0, 0, \ldots, 0\big), \qquad i = 1, \ldots, m,$$

and write $t_i = (i - 1/2)/m$. The following table lists the exact terminal widths on these records for every type of distinct two-coordinate batch:

queried coordinates	W_1	W_2
$\{t, z_j\}$	$H + mt_i - a_j$	$H + m(1-t_i) - b_j$
$\{t, u\}$	H	$H + m(1-t_i)$
$\{t, v\}$	$H + mt_i$	H
$\{u, v\}$	$H + M$	$H + M$
$\{u, z_j\}$	$H + M - a_j$	$2H - b_j$
$\{v, z_j\}$	$2H - a_j$	$H + M - b_j$
$\{z_j, z_k\},\ j \ne k$	$2H - a_j - a_k$	$2H - b_j - b_k$

For the first row,

$$\max(W_1, W_2) = H + |i - j| - \tfrac{3}{4},$$

so the batch succeeds exactly at x_j. The next two rows fail because $0 < t_i < 1$, and the fourth fails because $M > 0$. For the fifth and sixth, $a_j, b_j \le m + 1/4 < H$, so the displayed $2H$ term exceeds H. Finally,

$$(a_j + a_k) + (b_j + b_k) = 2m + 3 < 2H \qquad (m \ge 3).$$

At least one of $a_j + a_k, b_j + b_k$ is therefore below H, so every two-z batch fails as well. Thus each deterministic batch succeeds on at most $1/m$ of the adverse mass. The same holds for any batch mixture; its minimum success over records cannot exceed this adverse average.

Fewer than two distinct queries, repeated queries, and stopping early do not evade the bound. Extend any such query set to a distinct two-query set. Additional observations cannot enlarge either sharp interval, so the original success set is contained in the extended success set. The table therefore also bounds every permitted shorter execution. This completes the matching upper and lower bounds.

Finally, the fixed batch $\{t, z_m\}$ makes $W_1 \le H - 1/4$ for every record, and $\{t, z_1\}$ makes $W_2 \le H - 1/4$. The difficulty concerns serving both targets with the same two queries, not either target alone. □

Corollary 1 (All pairs of three models, threshold margin, and query cost)**.** *The construction can be realized as all three paired squared-loss comparisons among three nonnegative prediction vectors. The exact success values remain* 1 *and* $1/m$ *at every common normalized threshold*

$$\tau \in \left[\frac{1}{2} - \frac{1}{8H}, \frac{1}{2} + \frac{1}{8H}\right).$$

Each target separately remains fixed-batch certifiable, and three fixed queries suffice jointly throughout this interval. Hence the unbounded ratio concerns success probabilities at a common two-query budget, not an unbounded reduction in the number of readings.

Proof. Add $T_3 = T_1 - T_2$ and put

$$d_j = a_j - b_j = 2j - m - 1, \qquad P = \sum_{d_j > 0} d_j = \lfloor m^2/4 \rfloor.$$

The core $M(2t-1) + m(u-v)$ ranges from $-H$ to H, independently of the z box, whose contribution ranges from $-P$ to P. Thus T_3 has initial width $2(H+P)$. After querying t alone its exact width is $m + 2P$. Moreover,

$$(H+P)\left(1 - \frac{1}{4H}\right) - (m+2P) = M - P - \frac{1}{4}\left(1 + \frac{P}{H}\right) > 0,$$

because $M - P \geq m(m-1)/4 \geq 3/2$ and $P \leq H$. The third target therefore meets even the smallest displayed normalized threshold after the first query.

For the first two targets, the displayed interval is equivalent to an absolute width threshold $q = 2H\tau \in [H - 1/4, H + 1/4)$. The adaptive branches and uniform batch mixture retain their lower guarantees. On the adverse records, off-diagonal $\{t, z_j\}$ batches have maximum width at least $H + 1/4$. Every other table row has a larger failing width: at least $H + 1/2$ for $\{t, u\}, \{t, v\}$; $H + M$ for $\{u, v\}$; $H + M - 1/4$ for the mixed $u/v, z$ rows; and $H + M - 3/2$ for two-z batches. All exceed $H + 1/4$ when $m \geq 3$. Thus the same adverse distribution proves the upper bound $1/m$ throughout the stated half-open interval, also when success includes T_3. The upper endpoint is excluded because an off-diagonal adjacent batch then succeeds at equality.

The scalar batches from the theorem remain valid throughout the interval, and T_3 needs only t. The fixed three-query batch $\{t, z_1, z_m\}$ gives both first widths at most $H - 3/2$, while its third width cannot exceed the width after t alone. Thus three fixed queries suffice; two fixed queries cannot suffice because their optimal randomized success is only $1/m < 1$.

For an explicit model comparison, write $N = m + 3$ and choose the three nonnegative prediction vectors, in coordinate order $(t, u, v, z_1, \ldots, z_m)$,

$$p = (NM/2, 0, 0, 0, \ldots, 0),$$

$$q^{(1)} = (NM, Nm/2, 0, Na_1/2, \ldots, Na_m/2), \qquad q^{(2)} = (0, 0, Nm/2, Nb_1/2, \ldots, Nb_m/2).$$

Then $\mathrm{MSE}(p, x) - \mathrm{MSE}(q^{(r)}, x)$ has the coefficient vector of T_r, for $r = 1, 2$. Additive constants differ but do not affect widths. The remaining model-pair contrast has the coefficient vector of $T_1 - T_2$, up to orientation. Therefore the guarantee holds for all three comparisons, each normalized by its own initial width. □

The adaptive normalized margin at half-width is $1/(8H) = 1/[2m(2m+3)]$, which shrinks with m. These are exact-query constructed instances, not a fixed-error robustness result or a new empirical advantage on the cumulative-meter data. The scalar sparse-cover theorem concerns one affine target. Here each conditional joint width is the maximum of two affine functions; its strict failure set can be disconnected, so the scalar Helly argument does not supply a half-success floor.

The optimal mixture is unique. In the family of Theorem 2, any randomized batch with positive worst-record success must assign positive probability to every batch $\{t, z_j\}$, $j = 1, \ldots, m$. The unique optimal distribution over preselected unordered batches assigns probability $1/m$ to each of these m batches. Both statements also hold for all three model comparisons and every threshold in Corollary 1. To see this, at the adverse record x_i, only $\{t, z_i\}$ succeeds. Thus success there equals its mixture probability p_i. A positive worst-record value requires every $p_i > 0$; achieving $1/m$ requires every $p_i \geq 1/m$. Since the probabilities sum to at most one, all equal $1/m$ and no mass remains for other sets. The uniform mixture already provides the matching guarantee on every feasible record. Repeated or fewer queries cannot succeed at any x_i. Early stopping cannot improve the success of a preselected batch, so it does not evade the necessary support or mixture weights. Thus fixed output dimension two and a scalar first answer do not impose a universal bound on the number of batches needed for positive joint assurance.

Executable checks. Online Resource 1 includes a standard-library rational verifier and a separate LP calculation for $m = 3, 5, 10$. The rational checks enumerate 1,171 adverse-record/query-set combinations and verify continuum branch coverage. The 7,131 endpoint and whole-branch LPs agree with the exact formulas to at most 7.11×10^{-15}; the absolute comparison tolerance is 10^{-7} and the primal/dual feasibility tolerances are 10^{-9}. The proof above establishes the result for every $m \geq 3$; these finite checks test its implementation, including the third contrast and the threshold interval. From the extracted supplement root, run:

```
cd work/risk_adaptivity_revision/two_target_boundary
python -S verify_boundary.py --solver-free --output replay_exact.json
python verify_boundary.py --output replay_numeric.json
python -S verify_support.py --output replay_support.json
```

The first command needs only Python's standard library; the second also needs the pinned NumPy/SciPy environment. Both use the rational construction encoded in `verify_boundary.py`, with default sizes $m = 3, 5, 10$; no meter data are required. Successful reports contain `status: PASS`, `all_checks_passed: true`, input and script hashes, and the checked witnesses. The numerical report additionally records the 7,131 LPs and their discrepancies. These commands verify the construction; the complete source-to-table replay is documented in Supplementary Materials. The third command independently checks the necessary support and unique mixture on seven family sizes from $m = 3$ to 24, using 126,195 exact rational width evaluations, including the three-model and threshold extensions.

2.5 Shared precision from cumulative-meter readings

Consider a six-hour block with nonnegative hourly energy increments $z = (z_1, \ldots, z_6)$ and retained cumulative readings C_0, C_6. In common energy units, $E = C_6 - C_0$ gives

$F_E = \{z \geq 0 : \sum_h z_h = E\}$. For two frozen forecast vectors, the block MSE contrast is $D(z) = d + c^\top z$, with $c_h = 2(b_h - a_h)/6$. All six targets remain in its denominator.
The five optional queries return internal cumulative readings C_k, $k = 1, \ldots, 5$, equivalently exact prefix sums $s_k = C_k - C_0 = \sum_{h \leq k} z_h$. The budget is two distinct readings shared by every model comparison in the block. The endpoints are initially available and do not use this optional-query budget.
For $0 \leq p < q \leq 6$, put $r(p, q) = \max_{p<h\leq q} c_h - \min_{p<h\leq q} c_h$. Order the queried and outer boundaries as $0 = q_0 < \cdots < q_m = 6$, with $s_0 = 0$ and $s_6 = E$. The exact terminal width is

$$W_Q(z) = \sum_{\ell=1}^{m} (s_{q_\ell} - s_{q_{\ell-1}}) r(q_{\ell-1}, q_\ell). \tag{5}$$

Within each segment, its nonnegative energy can be placed at an hour with the smallest or largest coefficient. These choices attain both endpoints, proving sharpness. Initially $W_0 = Er(0, 6)$; zero-width contrasts require no further information. Online Resource 1, S6 retains the separate scalar closed forms and their controls. An auditor comparing several models can reuse each reading across their loss contrasts. Let $W_{r,Q}(y)$ be the sharp width for contrast r after the common transcript Q, and let $W_{r,0} > 0$ be its initial width. Define

$$J_Q(y) = \max_r W_{r,Q}(y)/W_{r,0}.$$

Then $J_Q(y) \leq 1/2$ means that every requested comparison has reached half its original width. Already determined contrasts can be omitted. This is the local diameter of the vector contrast map in a weighted ℓ_∞ norm, within the established vector-recovery framework (Krieg et al., 2025); shared-label model certification also has close prior work (Kuboyama, 2026).

2.5.1 Planning two shared prefix queries

We plan one pair of readings for all requested contrasts. The inputs are m normalized coefficient rows $c_r \in \mathbb{R}^n$, each with range one, an exact total $E > 0$, and all internal prefix queries $P_k x = \sum_{h \leq k} x_h$, where $x = y/E \geq 0$ and $\sum_h x_h = 1$. We assume $n \geq 3$ and a budget of two distinct queries. Define the range over a consecutive segment by $\Delta_r(a{:}b) = \max_{a\leq h\leq b} c_{rh} - \min_{a\leq h\leq b} c_{rh}$.
Suppose the first query is k and its answer is t. For a proposed second query $j \neq k$, the conditional worst width of contrast r is $\alpha_{kjr} t + \beta_{kjr}(1 - t)$, where

$$(\alpha_{kjr}, \beta_{kjr}) = \begin{cases} (\max\{\Delta_r(1{:}j), \Delta_r(j+1{:}k)\}, \Delta_r(k+1{:}n)), & j < k, \\ (\Delta_r(1{:}k), \max\{\Delta_r(k+1{:}j), \Delta_r(j+1{:}n)\}), & j > k. \end{cases}$$

Within each side of the first boundary, a worst-case allocation places its mass in the segment with the larger coefficient range. Maximizing over records and over the finite contrast family commutes, so

$$g_{kj}(t) = \max_r \{\alpha_{kjr} t + \beta_{kjr}(1 - t)\}, \qquad A = \min_k \max_{t\in[0,1]} \min_{j\neq k} g_{kj}(t)$$

is the optimal deterministic two-query worst-width guarantee.

Algorithm: two shared queries, exact-total simplex

1. For each first boundary k, form every affine line $\ell_{kjr}(t) = (\alpha_{kjr} - \beta_{kjr})t + \beta_{kjr}$.
2. Form B_k from 0, 1, and every intersection in $[0, 1]$ of two nonparallel lines, including lines belonging to different second queries.
3. Compute $A_k = \max_{t \in B_k} \min_{j \neq k} \max_r \ell_{kjr}(t)$. Choose the smallest boundary index attaining $\min_k A_k$.
4. Retrieve that boundary's counter value and calculate its normalized prefix mass t using the retained endpoint total.
5. Choose the smallest $j \neq k$ attaining $\min_j g_{kj}(t)$, retrieve it, and calculate all terminal segment-width certificates using the two shared answers.

Between consecutive points of B_k, the ordering of all underlying lines is fixed. The maximum within each group and then the minimum over groups therefore select one affine line on that interval. Its maximum is at an endpoint. This proves that the finite calculation covers every possible first answer; it is not a grid approximation. Equal or coincident lines are handled by the stated index rule.

The mathematical algorithm uses exact comparisons. The reported six-interval implementation uses the same construction in floating-point arithmetic: it treats values within 10^{-12} of a minimum as tied and chooses the smallest index, both offline and online. Its intersection routine skips slope differences of magnitude at most 10^{-14}; a 257-point grid is an additional numerical check. Exact rational branch and adversarial certificates support the reported exact assurance claims. The algorithm above addresses deterministic worst width on the uncapped, exact-total simplex. Bounded-error and capacity analyses use the separate conditional calculations described in their sensitivity protocols; randomized mean-width and precision-success comparisons retain their own optimization objectives.

2.5.2 Computing the minimum fixed reading budget

We can choose the reading budget instead of assuming it. Here $F = \{x \geq 0 : \sum_h x_h = 1\}$ and $\tau \geq 0$. For the normalized coefficients above, a fixed prefix set Q partitions the record into segments $\mathcal{S}(Q)$. Concentrating the unknown mass in one segment gives

$$\sup_{x \in F} J_Q(x) = \max_{r, S \in \mathcal{S}(Q)} \operatorname{range}(c_r|_S), \qquad b_{\text{fix}}(\tau) = \min_Q \{|Q| : \sup_{x \in F} J_Q(x) \leq \tau\}.$$

This yields a simple exact algorithm. Starting at the first interval, extend the current segment while every coefficient row has range at most τ. When the next interval would violate a bound, place a query immediately before it and start a new segment. The resulting partition has the fewest segments, hence the fewest readings: its first segment ends no earlier than that of any optimal partition, whose remaining segments can be truncated to give a valid partition of the remaining suffix. Induction completes the exchange argument. Maintaining each row's minimum and maximum takes $O(Rn)$ comparisons for R contrasts and n intervals.

Randomization cannot reduce this fixed budget when success one is required. Any positive-probability batch that fails on a feasible record prevents certainty at that record. We

compare the fixed minimum with the existing two-query adaptive certificate. If no single reading suffices and that certificate exists, the adaptive minimum is exactly two. Otherwise the minimum fixed batch supplies a constructive fallback; we do not optimize adaptive budgets above two. This calculation concerns exact, uncapped nonnegative allocations and noiseless prefix readings.

2.5.3 When two readings cannot guarantee precision

The planner also identifies an insufficient budget. Randomizing the adaptive choices cannot restore certainty in this case. The following finite-witness argument makes that conclusion checkable.

Proposition 2 (A certificate against randomized adaptive certainty)**.** *In the six-hour exact model with five permitted internal prefixes, suppose $A > \tau$. Then every randomized adaptive policy using at most two readings has worst-record precision success at most $19/20$. The bound need not be tight. Thus a policy with success one exists if and only if a deterministic adaptive policy with success one exists.*

Proof. For each of the five first queries k, choose an answer t_k at which every second query $j \neq k$ has $g_{kj}(t_k) > \tau$. Such an answer exists because $A > \tau$. For each ordered pair (k, j) choose a feasible record x_{kj} with first answer t_k on which the terminal sharp joint width exceeds τ. Put mass $1/20$ on these twenty records, retaining multiplicities. This adverse distribution is fixed before the policy's independent random seed is drawn.
Fix a seed. It determines a first query k and, upon answer t_k, a second query j. All four records in row k have that same first answer, so the policy follows this path on x_{kj} and fails. Its failure probability under the fixed adverse distribution is therefore at least $1/20$. Averaging over seeds preserves the bound. For each randomized policy there is therefore a fixed record with success at most $19/20$. Early stopping or repeated queries can be padded with unused legal queries; failure after padding implies failure before it because observations only shrink the compatible intervals. Conversely, a deterministic successful policy is a randomized policy with a constant seed. This proves the claim. □

For each x_{kj} the executable certificate supplies two nonnegative, unit-sum endpoint records with identical queried answers and a contrast difference strictly above τ. One endpoint serves as x_{kj}; the other proves failure of its sharp interval. The verifier checks these equalities and inequalities directly, without using an optimizer's verdict. This elementary finite-action argument concerns certainty under the stated model; it does not compute the optimal success probability below one.

Why the scalar precision bound does not apply to joint widths. Theorem 1 assumes one scalar target and an adaptive policy that guarantees precision. Its conditional worst widths are concave in the first answer, so their strict failure sets are convex. Here g_{kj} is a maximum of affine contrast widths and is convex; its failure set may be disconnected. The scalar Helly argument therefore does not transfer to joint precision. The joint advantage requires the matching certificates below. Figure 2(a) illustrates the conditional curves and the selected second query on one recorded task.

2.5.4 An inspectable certificate from the external panel

Consider HPM 1 on 5 July 2025, 12:00–18:00: the first matched-half case in the frozen external ledger. Normalize each of its six forecast contrasts to coefficient range one and write $x = y/E$ for the hourly allocation divided by its known block total. Thus $x \geq 0$, $\sum_h x_h = 1$, and joint precision means $J \leq 1/2$. The contrast order is daily–boosting, daily–MLP, daily–ridge, boosting–MLP, ridge–boosting, and ridge–MLP. The normalized coefficient rows have the following rational lower enclosures; every entry other than zero or one lies within 10^{-4} above its displayed value, and zero and one are exact:

$$C \in L + [0, 10^{-4}) \quad \text{entrywise}, \qquad L = \begin{pmatrix} 1 & .2928 & .0571 & .1475 & 0 & .0026 \\ 1 & .3251 & .6303 & .4169 & .1256 & 0 \\ 1 & .1466 & .0899 & .0242 & 0 & 0 \\ 0 & .3884 & 1 & .6778 & .6220 & .5040 \\ 1 & .4847 & .0140 & .3095 & 0 & .0061 \\ .2093 & .3546 & 1 & .7180 & .2280 & 0 \end{pmatrix}.$$

With $Q = 12$ and $R = 24$, their normalized width-coefficient vectors have the form

$$v_{Qr} = (0, 0, a_r, a_r, a_r, a_r), \qquad v_{Rr} = (b_r, b_r, d_r, d_r, e_r, e_r).$$

The following rational upper enclosures are obtained from the exact saved coefficients; each actual range lies between the displayed value minus 10^{-4} and that value, with zero and one exact.

Contrast r	$a_r \leq$	$b_r \leq$	$d_r \leq$	$e_r \leq$
1	0.1476	0.7072	0.0905	0.0027
2	0.6304	0.6749	0.2134	0.1257
3	0.0900	0.8534	0.0657	0
4	0.4960	0.3885	0.3222	0.1180
5	0.3096	0.5153	0.2955	0.0062
6	1	0.1453	0.2820	0.2281

Batch lower guarantee. For each contrast pair (r, s) choose

$$\lambda_{rs} = \begin{cases} 1, & r \in \{1, 3, 4, 5\}, \\ 1/2, & r = 2, \\ (1/3, 1/3, 3/7, 0, 1/4, 0)_s, & r = 6. \end{cases}$$

The table directly verifies all 36 conditions

$$\max\{(1 - \lambda_{rs})b_s, \lambda_{rs}a_r + (1 - \lambda_{rs})d_s, \lambda_{rs}a_r + (1 - \lambda_{rs})e_s\} \leq 1/2.$$

Thus Q and R cannot both fail: a failing contrast from each would make the indicated convex combination exceed $1/2$. Their equal mixture therefore succeeds with probability at least $1/2$ for every feasible record.

Matching upper bound. Place half the adverse mass at e_2 and half at $(e_4 + e_5)/2$. Batch 12 has joint widths zero and one there. Every other two-query batch leaves coordinate 2 in a segment containing coordinate 1 or 3. The saved coefficients satisfy $|c_1^{(3)} - c_2^{(3)}| \geq 0.8533$ and $|c_2^{(6)} - c_3^{(6)}| \geq 0.6453$, so all other batches fail at e_2. Every batch therefore succeeds on at most half of this adverse mass; every randomized batch has worst-case success at most $1/2$. The full-continuum batch optimum is exactly $1/2$.

Adaptive certificate and verification. The checked adaptive policy queries boundary 2, then 4 below its switch $\theta \in [0.5226, 0.5227]$ and 1 above it. The table bounds the two enlarged branch domains by 0.47743079 and 0.4774, respectively. These conservative bounds exceed the exact envelope peak, about 0.47734, plotted in Figure 2(a). Hence its success is guaranteed, even allowing either action inside the displayed switch enclosure. From the extracted package root, run the following commands; they require no optimization solver:

```
cd work/risk_adaptivity_revision
python joint_certification/proof_check_joint.py
python joint_certification/precision_batch_check.py
python joint_certification/precision_adaptive_link.py
python joint_certification/check_main_inspectable_certificate.py
```

The first three verify the constructed theorem and all 57 matching half-success certificates within the complete 279-block external panel. The last verifies this example's source hashes, 36 coefficient enclosures, 24 range enclosures, 36 multipliers, 20 adverse widths, and both branch domains. Full coefficients and the rational switch remain in its output. The original panel's 165 certificates require the documented raw-data regeneration route; these default commands do not reconstruct absent original inputs.

Precision on the original loss scale. Table 1 translates the relative target into loss units for the HPM 1 task above, using its frozen forecasts and two queried registers. The six-target loss uses $c_h = 2(b_h - a_h)/6$, so the interval widths are in kWh^2; a 24-target normalization is not used. The targets are stored API-grid increments under the measurement assumptions in Table 2. Exact segment ranges and separately implemented endpoint LPs reproduce the absolute widths.

Model contrast	Initial $W_{r,0}$	Target $W_{r,0}/2$	Terminal W_r	$W_r/W_{r,0}$
Daily–Boosting	0.303143	0.151571	0.001947	0.0064
Daily–MLP	0.194503	0.097252	0.005336	0.0274
Daily–Ridge	0.172075	0.086037	0.000673	0.0039
Boosting–MLP	0.213920	0.106960	0.004617	0.0216
Ridge–Boosting	0.131068	0.065534	0.001766	0.0135
Ridge–MLP	0.107125	0.053563	0.004663	0.0435

Table 1: Original loss scale for the HPM 1 worked example on 5 July 2025, 12:00–18:00 UTC. The first three numeric columns have units kWh^2: they are widths and width targets for identification intervals of six-target MSE differences on stored API-grid increments. The retained endpoint difference is $E = 1.83984375$ kWh. The adaptive policy retrieves the 14:00 register and then the 13:00 register. Normalization uses the initial width of each contrast, not a common numerical loss scale.

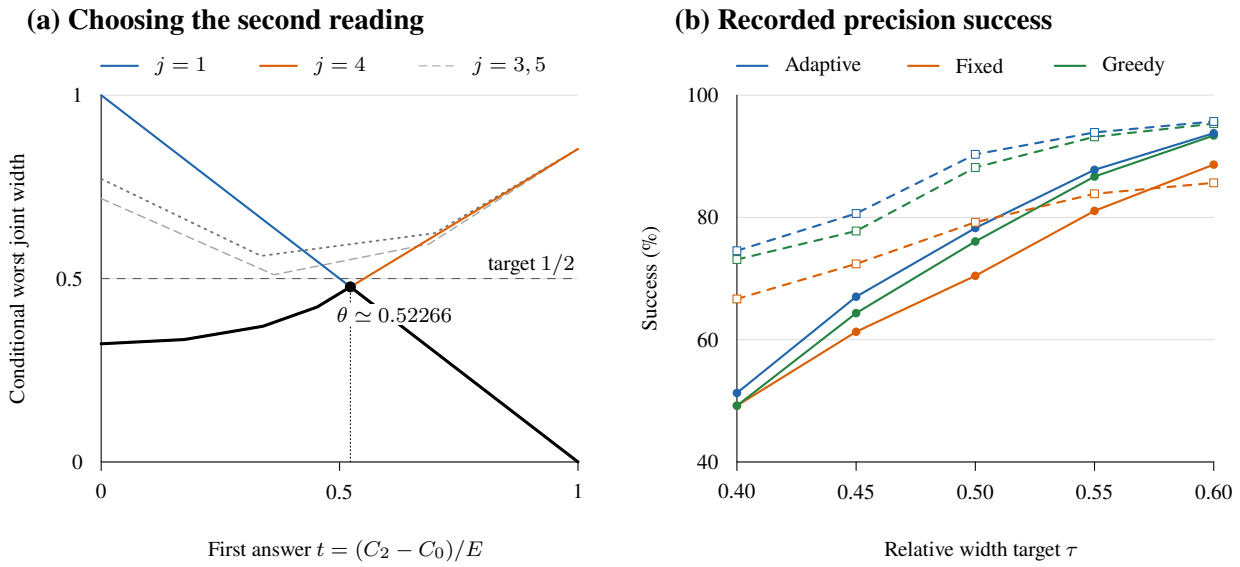


Figure 2: An acquisition rule and its recorded outcomes. (a) Exact conditional envelopes from the saved HPM 1 comparison on 5 July 2025, 12:00–18:00 UTC, after querying boundary 2. Gray curves show the other second-query options; the black lower envelope selects boundary 4 below the switch and boundary 1 above it. Its maximum is about 0.47734, below the half-width target for every possible first answer. (b) The adaptive, fixed-batch and greedy policies on all 819 original and 279 external blocks at five reporting targets. Solid curves represent the original panel and dashed curves the external panel. These exploratory recorded success rates use the existing 10^{-8} comparison tolerance; they are not worst-case precision probabilities.

2.6 Source panels and audit protocol

We use the first three consumer IDs in the pinned BLEM cumulative-counter source (Kostmann and Härdle, 2019). Training covers January–September 2017 and testing October–December 2017. Complete-day and feature filters leave 91 test days per meter. Hourly energy uses integer counter differences at exact UTC endpoints, converted to kWh without interpolation; partial days are excluded. Online Resource 1, S3 records source hashes and fitting details.

The external panel uses HeatpumpMonitor systems 1, 2 and 7 (OpenEnergyMonitor, 2026a,b). We screened ascending IDs for public MID status, Class-1 electrical-meter metadata and cumulative coverage of January–July 2025 before hourly retrieval. Training-only gates, fixed before July retrieval, required 95% valid hourly intervals and 90 complete training days. All three passed without replacement. January–June supplied training; July supplied 31 test days per system. We convert stored API-grid increments to kWh. Native acquisition and historical delivery times are unavailable (OpenEnergyMonitor, 2026c); S6 records the source-selection chronology. Both panels reuse frozen daily-persistence, ridge, boosted-tree and MLP forecasts. Each external model specification was fitted once without retuning. We audit blocks ending at 06:00, 12:00 and 18:00 UTC, giving 819 original and 279 external blocks. At each declared cutoff, all policies receive the endpoints and day-start forecasts; the following day's persistence copies are not yet available under this benchmark schedule. Two internal readings serve all six pairs of the four models. This is a benchmark access contract, not a verified restriction of either source's service; the public-API experiment also evaluates unrestricted bulk retrieval. The shared-query analysis is retrospective on both panels. Policies use frozen forecasts and, for adaptation, previously queried answers. We open withheld readings only for replay and require $W_r \leq W_{r,0}/2$ for every pair, using the uncapped exact-total initial widths throughout the shared-query sensitivity controls.

What the measurement model certifies. Table 2 separates stored-record guarantees from claims about physical energy. In each mathematical model, the guarantee requires the unknown record to satisfy the declared constraints.

Unchanged precision under capacity controls. For each shared contrast r, $W_{r,0}$ remains its initial width under the original uncapped exact-total model. The shared cap controls retain $W_r \leq W_{r,0}/2$: they test the same absolute uncertainty budget after adding hourly information. They do not test halving the new capped initial width. The scalar control in Online Resource 1, S6 reports both questions separately. We use the half-width fraction as a declared audit requirement; it is not a calibrated threshold for a downstream decision.

3 Results

The application asks which admissible retrieval method meets a required precision. It separates information available before a query, guarantees over all compatible records, replay on recorded answers, and actual API measurements. All policies in each comparison share the same initial information and measurement model. Scalar readings, network requests and feedback rounds are different resources. The protocol must identify which resource is limited before comparing methods.

Table 2: Targets and observation assumptions in the application. Error allowances and hourly caps are sensitivity inputs, not verified device limits.

Model	Target and observations	Scope of the guarantee
Exact stored samples	Targets are increments of stored cumulative samples. Retained endpoints and queried internal samples are exact in this record model.	Certificates concern loss contrasts on the stored grid. They do not establish exact physical energy or historical delivery times.
Bounded internal errors	The latent record includes one fixed internal-error vector with $\|e_k\| \le \delta E$, where E is the retained block total. Endpoints, and hence the retained total, remain exact. Errors are fixed before policy randomization.	Certificates cover every record and error vector within these bounds. The 1% allowance is illustrative; it is not sampled noise or device calibration.
Physical energy	Actual energy and meter readings would require defensible bounds for internal measurements, endpoints and any physical capacity constraints.	We provide no verified physical calibration. Source meter-class metadata alone does not establish the pointwise error model or exact endpoints.

Table 3: Numerical conventions for the central shared-query analysis. Exact certificates and floating-point replay have distinct roles.

Calculation	Convention
Constructed and matched certificates	Rational inequalities on the supplied exact inputs; no outcome tolerance.
Planner ties and intersections	Treat values within 10^{-12} of a minimum as tied; use the smallest index. Skip slope differences of magnitude at most 10^{-14}.
Shared mean-batch LP	Primal and dual feasibility tolerances 10^{-9}; secondary selection allows primary objective slack 10^{-10}.
Full-panel floating-point replay	Classify normalized widths using $J \le \tau + 10^{-8}$.
Exact cover replay	Use exact rational $J \le \tau$; the exploratory five-threshold analysis retains both conventions and reports their boundary difference.
Printed worked-example coefficients	The 10^{-4} interval is a rational enclosure of displayed coefficients, not a solver feasibility tolerance. Full coefficients drive the checks.

3.1 Joint precision with two shared readings

Table 4 separates the planning objectives from recorded precision success. Table 5 identifies where the precision challenger has matching bounds.

Policy	Planning objective	Guarantee and recorded comparison
Exact adaptive	Minimize worst joint width	Optimal among deterministic two-query policies in the declared exact model; replay reports joint precision success.
Fixed batch	Minimize worst joint width	Best deterministic two-query batch for that width objective; not optimized for recorded success.
Greedy adaptive	Myopic first query; optimal continuation	Optimizes the second query after observing the first answer; the first choice need not minimize final worst width.
Randomized mean batch	Minimize worst expected joint width	Batch-optimal for the uncapped exact simplex; replay reports expected width and precision success separately.
Cover + mean fallback	Certify batch precision success	Globally optimal where lower and upper certificates match; elsewhere uses the mean-width policy, with the remaining probability gap reported.

Table 4: Objectives are distinct even when all policies share the same two-reading budget, information and query dictionary. Recorded precision success always means $J \leq 1/2$; it is neither a minimax success value nor proof that a model winner is identified. The noisy adaptive control freezes an exact-model policy and certifies its guarantees anew, without asserting noisy-policy optimality.

Panel	Cases	Closed at 0	Closed at $1/2$	Closed at 1	Open gap	Fallback
Original	819	141	165	162	351	492
External	279	27	57	128	67	94

Table 5: Full-panel status of the batch-precision certificates. "Closed" means equal certified lower and upper bounds on the continuum batch precision optimum. The three closed columns and open-gap column partition each panel. Fallback is a separate, overlapping policy classification: it includes closed-zero and open-gap instances. Thus 468/819 and 212/279 gaps close, but only the 165 and 57 half-success cases provide the nontrivial 1-versus-$1/2$ assurance separation.

Shared-budget evaluation. Table 4 distinguishes the five acquisition policies. The "closed" and "open gap" labels in Table 5 describe whether the lower and upper precision bounds match. Every method receives the same totals, forecasts and two shared readings under Section 2.6. We set the half-width target and plan the policies before opening the withheld register responses. The

Panel / observation model	Blocks	Adaptive	Fixed batch	Greedy
Original, exact	819	641 (78.27%)	577 (70.45%)	623 (76.07%)
External, exact	279	252 (90.32%)	221 (79.21%)	246 (88.17%)
External, 1% allowance	279	242 (86.74%)	215 (77.06%)	235 (84.23%)

Table 6: Recorded-answer joint precision success with two shared readings. We select the fixed batch by its worst-width objective before answers; the last row reoptimizes this comparator for the error model while retaining the coefficient-derived adaptive and greedy policies. The 1% allowance enlarges compatible records around the stored answers with exact endpoints, as defined in Table 2.

greedy control first minimizes the worst one-query joint width, breaks ties using the best two-query continuation, and then minimizes conditional worst width after the first answer. The adaptive success gains over the best fixed batch for the worst-width objective are 7.81, 11.11 and 9.68 percentage points, respectively. Its gains over greedy are smaller: 2.20, 2.15 and 2.51 points. For the exact external panel, the mean realized J is 0.2823 for adaptive, 0.3381 for the fixed batch and 0.2893 for greedy. At the 1% allowance these are 0.3028, 0.3481 and 0.3099. The next comparison uses a batch policy designed for precision.

The cover + mean fallback policy uses a guaranteeing fixed batch when available, then a certified equal mixture of two batches, and otherwise a mean-width fallback. Its recorded success is 54.98% over the original panel and 72.55% over the external panel. These are replay rates for the declared cover and fallback. Matching certificates prove a batch precision optimum of $1/2$ on 165 original and 57 external instances. On these instances, adaptive succeeds on every stored record, while the optimal batch mixtures succeed with expected frequencies 108/165 (65.45%) and 37/57 (64.91%). Coefficient certificates define these subsets before record replay. The complete fixed-batch and greedy comparisons in Table 6 remain the broader controls.

A stronger guarantee can score worse on stored records. Table 7 makes the distinction visible on the same certified subsets. The fixed worst-width batch succeeds more often on the stored answers than the optimal randomized cover, yet its worst-case precision success is zero. Indeed, a deterministic batch's worst-case indicator is either zero or one; a value of one would contradict the proved randomized optimum of one half. Feedback attains both certainty over compatible records and success on these recorded cases.

Table 7: Assurance and replay answer different questions on the matched-half subsets. Randomized entries are expected success counts for the saved optimal mixtures. All methods use the same two readings and information.

Policy	Worst-case success	Original (165)	External (57)
Adaptive	1	165	57
Fixed worst-width batch	0	155	51
Optimal randomized cover	$1/2$	108	37

Capacity control. For an illustrative cap $y_h \leq E/2$, we reoptimize the noiseless planner using concave piecewise-linear segment widths. We solve each conditional problem by checking all line intersections within each capacity interval. Of the 819 original and 279 external records, respectively 780 and 200 satisfy this cap. Among those compatible records, adaptive-only half-width guarantees remain in 56 and 4 blocks; recorded adaptive versus fixed-batch successes are 655 versus 627 and 179 versus 177. Under the stronger cap $E/3$, only 613 original and 198 external records are compatible. No adaptive-only half-width guarantee remains, and the two planners have equal success counts of 613 and 192. We restrict cap-based guarantees to compatible records. Thus supplied capacity information can remove the need for feedback. Combining the $E/2$ cap with the 1% stored-reading allowance, the capacity-aware frozen adaptive policy succeeds on 169/200 compatible records, versus 166/200 for the best fixed batch under that model. Mean joint widths are 0.3479 and 0.3535. The greedy control succeeds on 165/200 using its frozen uncapped policy; it has not been reoptimized for the capacity model.

Paired uncertainty and source variation. The external exact-panel advantage over the worst-width fixed batch is 11.11 percentage points (descriptive 95% interval 8.60–13.62), compared with 2.15 points over greedy (1.08–3.23). These percentile intervals use 2,000 seven-day circular calendar-block resamples, synchronized across the three systems and intraday blocks, with seed 20261013. The combined cap/error gain over the fixed batch is 1.50 points (-0.49–3.16). Paired 2×2 counts, individual-system effects and one-day-cluster sensitivity appear in Online Resource 1, S8. The intervals describe these panels; three systems cannot establish population-level generalization, and dependence beyond the resampled blocks remains unmodeled. Table 8 shows the system-level paired counts. The external gain over fixed batching is

System	n	A / F / G	A-only/F-only	Δ_F (pp)	A-only/G-only	Δ_G (pp)
O1	273	212/193/211	27/8	6.96	3/2	0.37
O2	273	196/187/188	30/21	3.30	9/1	2.93
O3	273	233/197/224	38/2	13.19	12/3	3.30
HPM 1	93	76/75/73	3/2	1.08	3/0	3.23
HPM 2	93	91/66/87	25/0	26.88	4/0	4.30
HPM 7	93	85/80/86	5/0	5.38	0/1	-1.08

Table 8: Exact-reading paired joint-precision outcomes by system. A is the exact deterministic adaptive planner; F is the best deterministic batch for worst width; G is greedy adaptation. The two discordant counts identify blocks where only A or only its comparator succeeds. O1–O3 are BLEM consumers 1–3; HPM labels retain provider system identifiers. Blocks are repeated observations within systems.

heterogeneous: system 2 supplies 25 of the 31 net additional successes, while system 7 has one fewer success than greedy adaptation. Omitting each system in turn leaves positive exact-model panel effects: 5.13–10.07 and 3.23–16.13 percentage points over fixed batching for the original and external panels, respectively; the corresponding ranges over greedy are 1.65–3.11 and 1.08–3.76 points. A fourteen-day synchronized circular-block sensitivity also retains positive exact-model intervals (Online Resource 1, S8). These checks reuse the same observations; they are not additional independent replications. The primary seven-day cap-plus-error interval

remains $[-0.49, 3.16]$ points. Its fourteen-day sensitivity is $[0.48, 2.63]$, and its leave-one-system-out point effects range from 0.81 to 2.63 points. This small three-success gain remains sensitive to the temporal specification; the favorable secondary interval does not replace the primary assessment.

Checking batch-precision guarantees. For each selected batch pair, 36 rational multiplier checks prove that the two batches cannot both fail on any feasible record. Feasible adverse records give matching upper bounds, establishing the full-continuum $1/2$ optima. Rational checks also verify the adaptive half-width guarantees. Elsewhere, we report the gap between the challenger's lower and upper bounds; the mean-width fallback has only the trivial precision lower bound zero. Online Resource 1, S8 gives all multipliers, adverse completions and coefficient-based tie rules. These certificates assume exact readings; the next calculation treats errors separately.

When feedback changes the available guarantee. Exact rational checks partition the original 819 blocks into 162 where a fixed batch guarantees half-width, 168 where adaptive querying does so but no deterministic batch does, and 489 where no deterministic two-query policy does. The corresponding external counts are 128, 57 and 94. These are guarantees over the declared uncapped, noiseless simplex, not frequencies of success on the recorded trace.
Among the 165 original and 57 external blocks with matched randomized-batch optimum $1/2$, the greedy first query fails the half-width requirement on 25 and 13 blocks, respectively, even with its optimal conditional second query. Greedy nevertheless succeeds on the recorded trace in 20 of those 25 and 11 of those 13 blocks. The diagnostic identifies a specific benefit of planning the first query: recorded success can conceal an inadequate pre-query guarantee. Across the full panels, this first-query limitation occurs on 28 and 16 adaptive-guaranteed blocks. The supplied interval covers and uncovered-answer witnesses use rational arithmetic on the stored contrast coefficients.

A stronger fallback check. We retain the certified deterministic or equal two-batch cover and replace its mean-width fallback by the fixed worst-width batch on the same 492 original and 94 external fallback blocks. Using the strict threshold $J \leq 1/2$, this coefficient-only change raises recorded success from 54.98% and 72.55% to 64.71% and 74.19%.
Pure fixed batching still performs better (70.45% and 79.21%); the entire recorded-score deficit arises on the matched-half subsets, where fixed batching succeeds on 155/165 and 51/57 records, versus expected cover successes of 108/165 and 37/57. On each of those instances, however, every fixed batch has worst-case precision success zero, whereas the cover guarantees $1/2$. The matched-half guarantees and open continuum gaps also apply to this variant. This retrospective diagnostic reuses both panels; the supplement gives both threshold conventions, block-level paired outcomes, system summaries and descriptive seven-day block-bootstrap intervals.

Certifying an insufficient budget. Proposition 2 closes the last branch of Figure 1. All 489 original and 94 external blocks without a deterministic guarantee have twenty-record certificates bounding every randomized adaptive policy's success by $19/20 < 1$. A rational checker reconstructed all 6,588 source coefficient rows and 5,490 first-query covers, then verified 11,660 failing endpoint pairs. The smallest strict normalized failure margins were 0.00031746 and 0.00073903, respectively. These results apply to the uncapped, noiseless model. They justify

escalating an insufficient two-reading contract instead of hoping randomization will provide certainty. Online Resource 1 includes every synthetic witness; its default command checks the external panel, and the pinned raw-source route checks both panels.

How many readings does assurance require? The minimum fixed budget is three on every block where two adaptive readings suffice but two fixed readings do not: 168 original and 57 external blocks, spanning all six systems. Thus two adaptive readings replace three necessary and sufficient fixed readings on 225 blocks, a one-third reduction. Each replacement saves one reading in exchange for an additional information round: the three fixed readings can be chosen together. The original count includes the 165 matched-half cases and three further blocks; only the matched cases establish exact batch success one half. Using adaptive two on these blocks and the minimum fixed batch elsewhere gives the complete-panel reading totals in Table 9.

Table 9: Reserved optional scalar readings for guaranteed joint half-width precision. Each plan succeeds for every completion in the declared exact model. Counts exclude the two common retained endpoints. The adaptive plan uses a fixed fallback where no two-reading guarantee is available.

Quantity	Original	External
Blocks	819	279
Three fixed versus two adaptive	168	57
Minimum fixed readings, summed	2625	743
Adaptive with fixed fallback, summed	2457	686
Reading reduction across full panel	6.40%	7.67%

We calculated these budgets from the frozen coefficients before using any optional reading answers. Enumeration of all 32 fixed subsets per block agrees with the linear scan. A separately implemented shortest-path checker verifies minimality and checks feasible endpoint witnesses and the adaptive interval covers. Online Resource 1 supplies each concrete fallback batch, source and system summaries, and the fixed-budget histogram. The capacity and error controls test how additional information changes the acquisition requirement.

How much remains unresolved in the batch comparison. The batch-precision cover has matching continuum bounds on 468 of 819 original blocks and 212 of 279 external blocks. On the remaining 351 and 67 blocks, the median upper-minus-lower probability gap is respectively $1/3$ and $1/2$. The interquartile ranges are $[1/4, 1/2]$ and $[1/3, 1/2]$; both panels range from $1/5$ to $1/2$. Averaged over all blocks, including closed gaps, the gaps are 0.1583 and 0.0983. We use the 165 original and 57 external matched-half certificates as the evidence of exact precision superiority; the full-panel cover replay is a comparison with a declared policy, not with a globally solved precision optimum.

Sensitivity to the reporting target. Figure 2(b) gives an exploratory post-hoc replay of all 1,098 blocks at relative width targets $0.40, 0.45, 0.50, 0.55, 0.60$, keeping the query policies, fitted models and mixture parameters fixed. Across these targets, the adaptive policy's recorded

advantage over fixed batching ranges from 2.08 to 7.81 percentage points on the original panel and from 7.89 to 11.11 points on the external panel. The advantages over greedy range from 0.37 to 2.69 and from 0.36 to 2.87 points. Some descriptive seven-day-block intervals for the greedy comparisons include zero.

Shared-query sensitivity to stored-reading errors. Across all 279 external blocks, the frozen shared policy guarantees that all six normalized comparison widths are at most $1/2$ on 165 blocks under fixed internal-reading errors bounded by 1% of block energy; the best deterministic batch does so on 120. On the 45 additional blocks, feasible fixed-record certificates prove that every randomized batch has worst-case precision success below one. The certified batch success upper bound has minimum and median $1/2$, interquartile range $[1/2, 2/3]$, and maximum $4/5$; 26 bounds are at most $1/2$ and 37 are at most 0.7. These are upper bounds on the unrestricted batch optimum, not claims that each bound is attained. The 45 blocks span all three external systems (11, 11, and 23 blocks).

Capacity information reduces the benefit. With a per-hour cap of half the block energy as well as the same 1% error allowance, a policy frozen from the capacity-aware noiseless planner certifies 213 of the 279 coefficient instances, versus 204 for the best deterministic batch; nine have a certified randomized-batch limitation, with upper bounds between $2/3$ and $7/8$. Only 200 recorded profiles satisfy this illustrative cap. Within that compatible subset the counts are 153 versus 147, with six separations. The original uncapped policy certifies only 191 of the 279 capped instances, which motivates using declared capacity information when planning. Both policies are fixed before the fine readings are opened; neither is claimed optimal for bounded-error acquisition.

The comparison uses two shared readings, uncapped noiseless initial widths for normalization, exact block endpoints, and the same permitted queries for every method. A separately implemented rational checker verifies the saved contrast coefficients, branch coverage, every reported precision status, and the complete fixed-error adverse records. Supplementary methods (Online Resource 1, S8) give the certificate construction.

Reproducibility and numerical checks. The supplement connects version-pinned source records, frozen forecasts, contrast coefficients, saved policies and reported outcomes. A fresh both-source run reproduced all 1,098 outcome rows exactly, including the 165 original and 57 external matched-half certificates. The original source route rejects mismatched files; its raw records are regenerated locally. Default offline replay uses the bundled inputs and is separate from raw regeneration and fresh model fitting.

A separately written checker reconstructs the policies and summaries. On 18 fixed representative blocks, 19,440 endpoint LPs and 6,480 conditional LPs have maximum discrepancies of 2.22×10^{-16} and 1.11×10^{-16}, against a 10^{-8} check tolerance. Exact rational covering and adverse-record inequalities establish the matched precision claims. An optimal-face diagnostic also checks that arbitrary selection among mean-optimal batch mixtures does not explain their lower recorded success. Online Resource 1, S8 supplies the full checks and the results-to-command map identifies each reproduction entry point.

3.2 What the retrieval interface changes

The HeatpumpMonitor API permits full-range retrieval, so we tested it directly (OpenEnergyMonitor, 2026b,c). We fixed the first eligible daily-persistence–MLP task from each external system before querying its optional answers. Two repetitions of five policies produced 30 trials and 60 HTTP requests, including common endpoint reads.

Table 10: Observed public-API control: three tasks, two repetitions per policy. Incremental I/O excludes client throttle waiting. The full-range response contains seven hourly samples, including five internal readings. All five tested policies met the precision target; the full five-policy measurements and cached response hashes are in Online Resource 1, S15.

Policy	Requests	Mean I/O (s)	Mean body bytes
Exact adaptive	2	0.497	104.3
Full hourly range	1	0.331	226.7

Bulk retrieval uses one request, achieves zero terminal width and has lower observed I/O time on these tasks. We therefore recommend bulk when the full response is admissible. For selective access, the minimum-budget calculation identifies the readings needed for assurance and the possible reduction from feedback. Online Resource 1, S8 supplies the protocol selector and its reading and acquisition-round cost catalogue.

4 Discussion

4.1 Planning the readings needed for assurance

The auditor supplies the comparisons, available information and permissible measurements. Our method returns the minimum sufficient fixed reading budget and checks whether feedback reduces it. On 168 original and 57 external blocks, three fixed readings are necessary and sufficient, while two adaptive readings suffice. Each saving exchanges one scalar reading for an additional information round. This calculation lets the auditor compare plans before opening the optional measurements.

Theorem 2 explains why sharing readings matters. Two simultaneous comparisons can remove the scalar two-query success floor: adaptive certainty coexists with optimal batch success $1/m$, although each contrast separately admits a sufficient fixed batch. The adverse records force all m batches even for positive worst-record success and identify the unique optimal mixture. The output dimension and query budget stay fixed as the dictionary grows; three fixed readings suffice in this construction.

The meter study connects this distinction to frozen forecast geometries. With two readings shared across six comparisons, 165 original and 57 external blocks have adaptive success one and optimal batch success one half. The worked example reports both normalized widths and original loss units (Table 1). On recorded answers, joint precision gains are 7.81 and 11.11 percentage points over fixed worst-width batching and 2.20 and 2.15 over greedy adaptation. Separate-pair gains are 0.407 and 0.119 points.

4.2 Scope and protocol choice

These guarantees apply to the specified information and query dictionary. The main planner uses two cumulative-prefix queries and an exact nonnegative total. The minimum-budget result counts scalar readings; the API control measures requests and I/O time. Bulk retrieval wins on that three-task workload: one request returns the complete record faster and leaves zero width. A selective plan is therefore relevant when the auditor limits individual readings; an admissible full-record export favors bulk. The source panels do not establish a deployed selective-access service.

Additional information can change the plan. The stronger $E/3$ capacity control removes the adaptive-only half-width advantage on compatible records. Bounded errors can increase the sufficient reading budget. Table 2 states these sensitivity models, including exact retained endpoints and uncalibrated error allowances. The source panels each contain three systems and were reused retrospectively. Matching certificates establish the exact batch optima on the reported subsets; 351 original and 67 external full-panel precision gaps remain open.

Precision is a reporting requirement; model selection also depends on the intervals' positions. On the external panel, adaptation certifies a best model on 207 blocks and greedy on 208. Original counts are 123 and 118; fixed batches certify 122 original and 201 external blocks. Thus the half-width criterion measures reliable comparison precision, with winner certification reported as a separate outcome. Online Resource 1 contains the broader scalar theory, coordinate-recovery solver and protocol-selection cost sensitivity. A next step is to develop scalable planners for larger adaptive reading budgets.

5 Conclusions

Sharing a reading budget can make feedback necessary for an assurance requirement that each comparison can meet separately. With two affine targets and two coordinate queries, adaptive success can be one while optimal randomized-batch success is $1/m$. The same construction covers all pairs among three models.

Our exact planner and verifiable certificates turn this boundary into a reading-budget decision. On 225 blocks across six systems, two adaptive readings replace the minimum three fixed readings; 222 also have matching batch-success bounds of one half. Combining these adaptive plans with minimum fixed batches elsewhere reduces reserved readings by 6.40% and 7.67% across the original and external panels under the exact-total model. The resulting plans specify the readings and feedback needed to attain a stated assurance level, allowing an auditor to select a protocol that fits the available access interface.

Supplementary Materials

Online Resource 1 is supplied as Supplemental Data S1 and S2. Extract both ZIP files into the same directory. Together they contain the editable manuscript, executable methods, frozen forecasts, source identifiers, case ledgers, exact witnesses and numerical checks. Start with `README.txt`; `CURRENT_RESULTS_MAP.txt` maps each current table and verification claim to its command, inputs and outputs. The default verification replays saved forecasts and does not train

new models. Separate commands cover raw-input regeneration and fresh fitting. BLEM-derived traces are reconstructed locally; attributed HeatpumpMonitor records are bundled. The indexed extended methods retain full proofs, solver scaling, historical controls, forecasting protocols and uncertainty checks. Protocol-selection files include all 116 interface decisions and task-bound readout checks. The separate checker compares saved contrasts with frozen geometry, verifies forecast hashes, and reconstructs retained endpoints from cached API records. Historical reports relevant to this revision are retained; the larger development archive is preserved separately.

Additional Information and Declarations

Data and Code Availability

The numerical code and detailed outputs are supplied in Online Resource 1. The public source identifiers are given in the references. The package separates bundled Portugal/household replay from BLEM raw-source regeneration. BLEM counter traces are regenerated from the pinned public QuantLet repository (`https://github.com/QuantLet/BLEM`), accompanying Kostmann and Härdle (2019; `https://doi.org/10.3390/en12142718`). The regeneration commands reject inputs with mismatched hashes. These raw traces are not redistributed because redistribution permission has not been established. Constructed polytopes include their complete inequalities and objectives. Dual outputs retain raw multipliers, any numerical mass correction, the completed record, and pricing and enumerated lower bounds. LP and endpoint cross-checks use floating-point arithmetic with explicit tolerances; the counter covering and adverse-profile certificates are additionally checked with exact rational arithmetic on their serialized inputs. Shared-query checks include all joint outcomes, optimal-batch tie diagnostics, bounded-error dual certificates and adverse completions. Reviewers receive the current executable materials as Supplemental Data S1 and S2.

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Competing Interests

The author declares no competing interests.

Author Contributions

Jishan Kapoor defined the research objectives and evaluation requirements, directed and supervised the analysis and revisions, interpreted the findings, and approved the manuscript. The computational and drafting assistance used under this supervision is described below.

Research Transparency

This work uses public data and constructed mathematical instances, with no newly recruited participants. New measurements concern public API response times and payloads, with their small-workload scope reported. Retrospective masks and archive-access assumptions are identified explicitly. Comparisons use the same information and the same stated hard budgets. The new proofs and numerical checks do not change the historical forecast results.

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